

Yoni Nachmany

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Engineer and leader with research experience and capacity for building technical systems supporting multimodal foundation models, multidimensional data pipelines, and deep learning applications in CV/NLP. Conference speaker.

Experience

Zeus AI

Remote

Head of Data Engineering

May 2024 – May 2025

Zeus AI spun out of NASA to develop ML models for data assimilation and forecasting with Earth observations. As the first hire, I built large-scale infrastructure and efficient workflows for generating training data and serving models.

- Operationalized PyTorch Lightning foundation model for gap-filled global atmospheric weather 50x faster than traditional systems based on Apple's 4M (Massively Multimodal Masked Modeling) applied to 4D sensor data.
- Architected live ingestion/inference pipelines with AWS S3/SNS/Lambda and H100s on Modal/Lambda Labs.
- Scraped, filtered, normalized, and tokenized TBs of video-like satellite imagery with tens of variables on HPC.
- Deployed web demo of hourly changes in solar potential for energy users, visualizing Zarr arrays in the browser.

The Linux Foundation

Contract, Remote

Senior Data Scientist

March 2024 – May 2024

The Overture Maps Foundation is an open map data collaboration under Linux with dozens of member companies. In the Data Pipeline Task Force, I contributed to the Global Entity Reference System for better data conflation/joining.

- Prototyped Parquet changelogs with PySpark/Databricks for map data used by Microsoft, Meta, and Amazon.

Gro Intelligence

New York City

Data Scientist, Project Lead

April 2023 – March 2024

Gro Intelligence built an agricultural data platform with predictive models mapping the impact of climate change on commodities. As Data Platform Project Lead, I managed the development of internal data computation/cataloging.

- Orchestrated 33 Airflow DAGs to compute 100+ years of past/projected indices in a versioned array data lake.
- Implemented and oversaw development of internal libraries to standardize Xarray metadata and input/output.

The New York Times

Contract, New York City

R&D Engineer

December 2021 – September 2022

The NYTimes R&D Team explores emerging technologies and develops new technical capabilities and story formats.

I contributed to 3D mapping research while owning the CV/NLP work to create a more accurate digital news archive.

- Scaled transcription of scanned newspapers with multimodal APIs and evaluation framework for 10M articles.
- Implemented 3D satellite stereo reconstruction with photogrammetry libraries to support newsroom coverage.

Mapbox

Washington, D.C.

Software Engineer

September 2019 – November 2021

Mapbox's Imagery team maintains and improves the satellite basemap by sourcing, updating, and enhancing provider data. I extended the capabilities of the data processing pipeline to handle new sources and derive value-added data.

- Updated 135M km² of satellite imagery for 3D map launch and saved \$1M+ by leveraging open aerial imagery.
- Improved the Mapbox Satellite UX for all users loading data across zoom levels groups with novel data fusion.
- Implemented and evaluated deep learning-based object detection/super-resolution (EDSR) for map customers.

Education

University of Pennsylvania

Philadelphia

MSE, Data Science

May 2019

BSE, Networked and Social Systems Engineering

May 2018

Hunter College High School

New York City

Keywords: deep learning, ML weather prediction, digital twins, multilingual AI, crowdsourcing, medical imagery.

Tech: Python, PyTorch, W&B, FastAPI, PySpark, Parquet, Airflow, Xarray, Dask, Zarr, Pandas, Mapbox GL JS.

Interests: art, architecture, movies, live music, running, travel, theater, Jewish learning, New York City history.

Research

<i>Global Atmospheric Data Assimilation with Multi-Modal Masked Autoencoders</i>	Preprint 2024
<i>Learning Word Ratings for Empathy and Distress from Document-Level User Responses</i>	LREC 2020
<i>Design Considerations for High Impact, Automated Echocardiogram Analysis</i>	ICML 2020
<i>Mitigating Geographic Bias of Image Classifiers with Multilingual Image Data</i>	D4GX 2019
<i>Detecting Roads from Satellite Imagery in the Developing World</i>	CVPR 2019
<i>Generating a Training Dataset for Land Cover Classification to Advance Global Development</i>	NeurIPS 2018
<i>Decoded: Crowdsourcing Summaries of Internet Privacy Policies</i>	HCOMP 2017

Talks

<i>Generating Elevation Data from Stereo Pairs of Satellite Imagery</i>	GeoPython 2023
<i>Depth Perceptions - Perspectives on 3D & Digital Twins</i>	SatSummit 2022
<i>Mapping Historical “Street View” Images of New York City: Visualizing Geotagged Archival Photos</i>	FOSS4G 2022
<i>Under-Mapped Spaces: New Methods and Tools for Critical Storytelling</i>	Stanford 2022
<i>Deep Transfer Learning for Land Cover Classification on Open Multispectral Satellite Imagery</i>	FOSS4G 2019

Teaching

University of Pennsylvania	Philadelphia
TA for NETS 213: Crowdsourcing & Human Computation with Prof. Chris Callison-Burch	Spring 2019
TA for ESE 305: Foundations of Data Science with Prof. Victor Preciado	Fall 2017